



Ji JiaHao (Martin Ji)

Inner Mongolia Autonomous Region, China · +1 (410) 746 8935 · jji30@jh.edu

EDUCATION

South China University of Technology – Guangzhou, China

Sep 2021 – Jul 2025

Bachelor of Engineering in Network Engineering

- IT Related Courses: *Introduction to Information Security / Digital System Creative Design / Foundations of Computer Science / Advanced Language Program Design (C++) I & II / Advanced Topics of Information Technology / Data Structures / Computer Organization & Architecture / Database System / Data Security for The Internet Plus / Computer Network / Operating Systems / Artificial Intelligence*
- Received University Scholarship and Outstanding Student Leader
- Community Service: Wetland Conservation Club; Sunflower Volunteer Association Member

Johns Hopkins University – Baltimore, MD, USA

Aug 2025 – Present

Master of Science in Security Informatics (MSSI)

- Research Interests: AI-driven cybersecurity, Web and Browser Security

RESEARCH EXPERIENCE

Amazon Nova AI Challenge, Trusted Software Agents Track

- Ongoing, results and technical details under program confidentiality.

AI-Based Multimodal Depression Detection Project – SCUT, Guangzhou, China

Aug 2024 – Feb 2025

Research Intern, Artificial Intelligence Laboratory

- Developed a model to detect signs of depression using multimodal data (3D video, audio, text). Applied mutual information maximization techniques to enhance feature representation across modalities.
- Used computer vision and natural language processing tech to study facial expressions, vocal qualities, and text transcripts of interviews. Created a Patient Health Questionnaire to diagnose depression.
- Used a customized mutual information maximization algorithm to capture shared representations between different data types and improve the model's ability to detect subtle indicators of depression.

Data-Driven Enterprise Intelligence and Organizational Decision Systems - Guangzhou, China

Dec 2023 – Aug 2024

- Conducted interdisciplinary research under Professor Wu, an associate professor at CUHK Business School and senior scientist at Hong Kong FinTech Laboratory, focusing on data-driven enterprise information systems and computational modeling for business intelligence.
- Investigated the intersection of geopolitics, supply chain resilience, and FinTech innovation, applying machine learning and big data mining techniques to analyze systemic risks and decision optimization in commerce.
- Published a paper, “The Application of Big Data Statistical Analysis Method in the Domain of Corporate Financial Management,” in the *International Journal of Global Economics and Management (IJGEM)*.

EEG-Based Brainwave Music Generation and Application Project – SCUT, Guangzhou, China

May 2022 – May 2025

Undergraduate Research Program (SRP)

- Designed a system that transforms EEG signals into generative music to reflect users' emotional states in real time.
- Preprocessed EEG data through band-pass filtering, artifact removal (ICA), and differential entropy(DE) feature extraction, following methods from DEAP dataset studies.
- Used emotion classification models (CNN + LSTM hybrid) to map brainwave features to arousal–valence dimensions.
- Integrated conditional generative adversarial networks (cGANs) to dynamically translate emotional states (arousal–valence) into MIDI control parameters for tempo, chord progression, and tonality.
- Implemented the full pipeline in Python (PyTorch, Librosa, and MNE), enabling both offline model training and real-time EEG-to-music streaming.
- Validated emotional congruence through SAM-based subjective ratings and spectral centroid–energy correlation analysis between EEG patterns and generated audio.

WORK EXPERIENCE

Guangdong Yixun Technology Co. Ltd – Guangzhou, China

Jul 2024 – Nov 2024

AI Algorithm Engineering Intern

- Assigned to work with China Telecom. Developed/implemented AI algorithms using the MindSpore framework & the YOLOv7 model to train urban governance AI systems and advance smart city technologies.
- Helped develop 20+ algorithms, including fire detection/fall detection/high-altitude parabola detection. Integrated these into city cameras and automated police-alert responses.
- Trained/fine-tuned the YOLOv7 model for object detection tasks.
- Integrated AI models into real-time monitoring systems and optimized inference pipelines to reduce latency, enable accurate frame-by-frame processing, and ensure timely detection of critical events.

Yan Jia Media Technology Co., Ltd. – Guangzhou, China

Aug 2023 – Oct 2023

A boutique software development firm focused on social media technologies

Lead Intern, Database Development Team

- Designed MySQL databases for optimized data storage. Maintained data integrity/security via regular upkeep, employing encryption protocols, access control mechanisms, and periodic security audits.
- Utilized Python/Excel/Tableau to analyze complex data sets to extract valuable insights into market trends, consumer behavior, and product performance. Used these insights to generate data-driven recommendations aimed at optimizing product designs, refining FY2024 marketing strategies, and enhancing overall operational efficiency.
- Enhanced database performance by optimizing query execution via index optimization. Resolved slow query response times by reconfiguring index structures, resulting in a notable decrease in overall latency.

Mingxin Technology Co., Ltd. – Baotao, Inner Mongolia, China

Dec 2022 – Jan 2023

A tech-driven enterprise specializing in the production of rare earth materials like Lanthanum (La) and Cerium (Ce)

Intern, Research and Development Department

- Used Python to automate rare earth production equipment. Leveraged Arduino IDE software to program microcontrollers.
- Used integrated sensors & actuators to monitor material levels and automatically alert workers to replenish inventory if necessary during the raw material preparation phase.
- Implemented programmable control modules to perform precise tasks during processing, ensuring consistent and accurate product quality.
- Helped design a powder-conveying device that was patented through the China National Intellectual Property Administration. *Patent ID: 202310184123.4*

ADDITIONAL INFORMATION

- IT Skill set: MySQL, Linux, C, C++, Java, Python, JavaScript, HTML5, CSS, Visual Studio, WebStorm, Pycharm, IDEA, Qt, MATLAB, Dev-C++, Wireshark